

Traivo Artikelvy: Hybrid Layout Implementation Task

Projektkod: TRAIVO-ARTICLE-HYBRID-v1
Target: Replit Implementation
Priority: High
Estimated effort: 3-5 dagar

1. Executive Summary

Problemet

Nuvarande artikelvy använder en **vertikal accordion-layout** som fungerar dåligt på stora skärmar (27"+):

- Användare måste scrolla långa sträckor mellan sektioner
- Översikten över artikelns status (vilka sektioner är ifyllda) saknas
- Ineffektiv användning av skärmutrymme - mycket tomrum på stora skärmar
- Svårt att snabbt hoppa mellan sektioner utan att scrolla
- Ingen visuell feedback på progress/completeness

Lösningen: Hybrid Layout

En **Sticky Quick Nav** sidebar kombinerad med **Smart Expansion** i huvudområdet:

Quick Nav (Sticky) 15%	Main Content Area 85%
✓ Grunddata	[Grunddata - Always Expanded]
▲ Lager	[Lager & Inköp - Collapsed with Badge]
○ Ekonomi	[Ekonomi - Collapsed]
...	...

Fördelar:

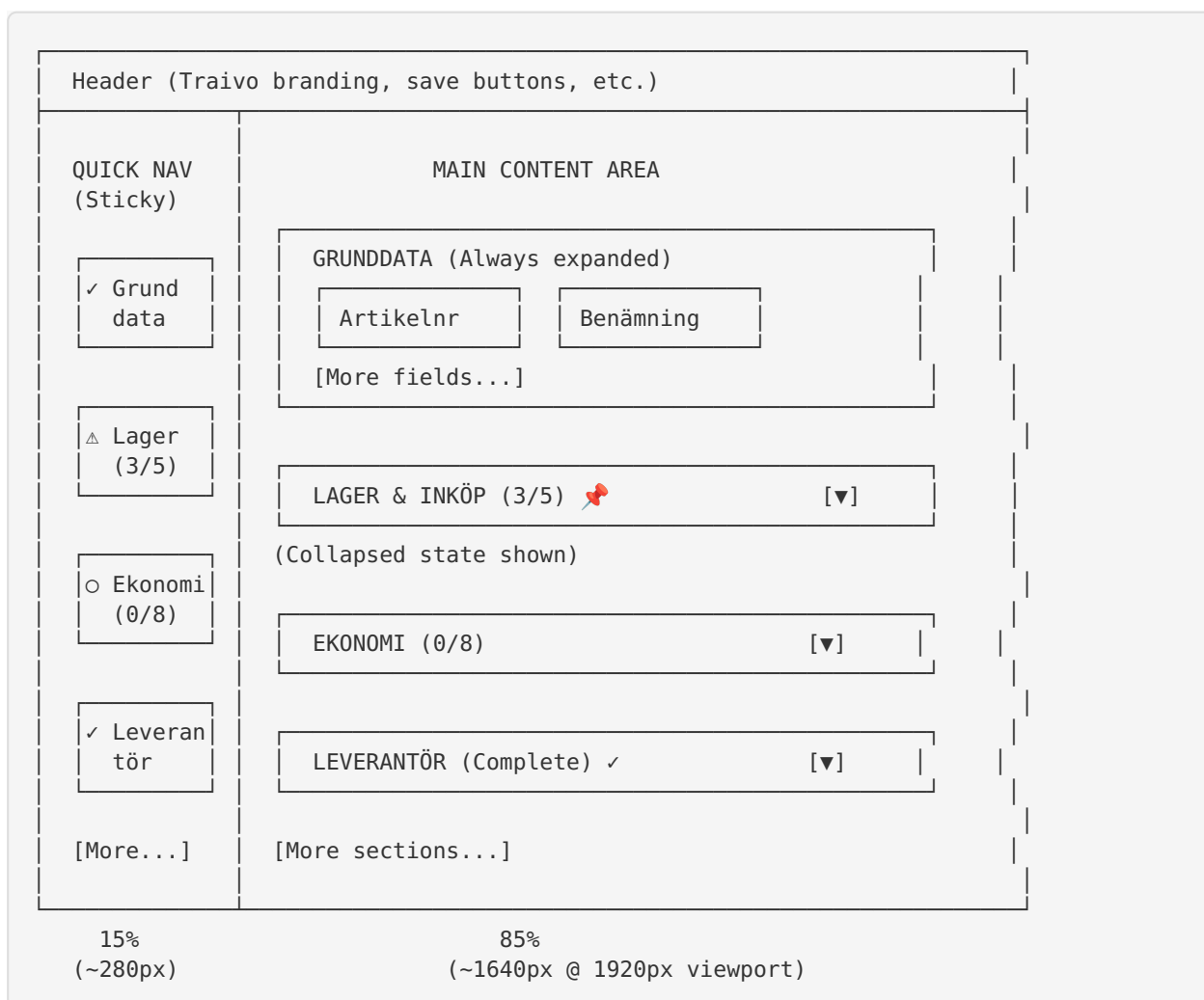
- **Översikt:** Alla sektioner synliga i sidebar med status-indikatorer
- **Snabb navigation:** Ett klick för att hoppa till och expandera sektion
- **Effektiv skärmanvändning:** Sidebar + main content använder hela bredden
- **Progress visibility:** Badges visar antal ifyllda fält per sektion
- **Personalisering:** Pin viktiga sektioner, state persistence

Förväntad effekt

- ⬇️ 60% reduktion i scrolling för vanliga uppgifter
- ⬆️ 40% snabbare att navigera mellan sektioner
- ⬆️ Ökad översikt över artikelns completeness
- ⬆️ Bättre användarupplevelse på stora skärmar

2. Visuell Layout-spec

Desktop Layout (≥ 1920 px bredd)



Exakta proportioner och spacing

Container:

- Total width: 100% viewport
- Max-width: none (fullscreen)
- Padding: 0 (sections handle their own padding)

Sidebar (Quick Nav):

- Width: 15% (min: 240px, max: 320px)
- Position: `position: sticky; top: 80px;` (80px för header height)
- Height: `calc(100vh - 80px)`
- Overflow-y: `auto`
- Background: `#F8F9FA`
- Border-right: `1px solid #E0E0E0`
- Padding: 24px 16px
- Z-index: 100

Main Content:

- Width: 85% (flex-grow för att ta resterande space)

- Padding: 32px 48px
- Background: #FFFFFF
- Max-width: 1400px (för att undvika för långa läsrader)

Spacing System:

```
--spacing-xs: 4px;
--spacing-sm: 8px;
--spacing-md: 16px;
--spacing-lg: 24px;
--spacing-xl: 32px;
--spacing-2xl: 48px;
```

Section Spacing:

- Margin bottom mellan sektioner: var(--spacing-lg) (24px)
- Padding inom sektion header: var(--spacing-md) (16px)
- Padding inom sektion content: var(--spacing-lg) (24px)

3. Sidebar Navigation (Quick Nav)

Struktur och innehåll

```
<QuickNav>
  <SectionLink
    icon="✓"
    label="Grunddata"
    status="complete"
    isActive={true}
    isPinned={false}
  />
  <SectionLink
    icon="△"
    label="Lager & Inköp"
    status="partial"
    badge="3/5"
    isActive={false}
    isPinned={true}
  />
  <SectionLink
    icon="○"
    label="Ekonomi"
    status="empty"
    badge="0/8"
    isActive={false}
    isPinned={false}
  />
  { /* ... more sections */ }
</QuickNav>
```

Visuella indikatorer

Status Icons:

- ✓ (Checkmark, grön) - Sektion komplett (alla required fields ifyllda)

- ⚠ (Warning, gul) - Sektion delvis ifylld (några fields har värden)
- ○ (Circle, grå) - Sektion tom (inga fields ifyllda)

Färgkodning:

```
.section-link--complete {
  border-left: 3px solid #4CAF50; /* Grön */
}

.section-link--partial {
  border-left: 3px solid #FF9800; /* Orange */
}

.section-link--empty {
  border-left: 3px solid #E0E0E0; /* Grå */
}

.section-link--active {
  background: #E3F2FD; /* Light blue */
  border-left: 3px solid #2196F3; /* Blue */
  font-weight: 600;
}
```

Badge System:

```
<span className="section-badge">3/5</span>
```

- Position: högerjusterad inom link
- Background: #E0E0E0 (tom), #FFE0B2 (partial), #C8E6C9 (complete)
- Font-size: 12px
- Padding: 2px 6px
- Border-radius: 12px

Pin Indicator:

```
{isPinned && <span className="pin-icon">📌</span>}
```

- Position: före sektionsnamnet
- Font-size: 14px

Sticky behavior

```
.quick-nav {  
  position: sticky;  
  top: 80px; /* Height of main header */  
  height: calc(100vh - 80px);  
  overflow-y: auto;  
  overflow-x: hidden;  
  
  /* Smooth scrollbar */  
  scrollbar-width: thin;  
  scrollbar-color: #BDBDBD #F8F9FA;  
}  
  
.quick-nav::-webkit-scrollbar {  
  width: 6px;  
}  
  
.quick-nav::-webkit-scrollbar-thumb {  
  background: #BDBDBD;  
  border-radius: 3px;  
}
```

Hover states och active states

```
/* Default state */
.section-link {
  display: flex;
  align-items: center;
  gap: 8px;
  padding: 12px 12px;
  margin-bottom: 4px;
  border-radius: 6px;
  border-left: 3px solid transparent;
  cursor: pointer;
  transition: all 0.2s ease;
  font-size: 14px;
  color: #424242;
}

/* Hover state */
.section-link:hover {
  background: #F5F5F5;
  transform: translateX(2px);
}

/* Active state (current section in view) */
.section-link--active {
  background: #E3F2FD;
  border-left: 3px solid #2196F3;
  font-weight: 600;
  color: #1976D2;
}

/* Focus state (keyboard navigation) */
.section-link:focus {
  outline: 2px solid #2196F3;
  outline-offset: 2px;
}

/* Pinned section styling */
.section-link--pinned {
  background: #FFF9C4;
  border-top: 1px solid #FFE082;
  border-bottom: 1px solid #FFE082;
}
```

4. Huvudområde (Main Content)

Grunddata: Alltid expanderad

```
<Section
  id="grunddata"
  title="Grunddata"
  alwaysExpanded={true}
  className="section--grunddata"
>
  <FieldGrid>
    <TextField label="Artikelnummer" name="artikelnr" required />
    <TextField label="Benämning" name="benamning" required />
    <TextField label="Beskrivning" name="beskrivning" />
    { /* ... more fields */ }
  </FieldGrid>
</Section>
```

Styling:

```
.section--grunddata {
  background: #FAFAFA;
  border: 2px solid #2196F3;
  border-radius: 8px;
  padding: 24px;
  margin-bottom: 24px;
}

.section--grunddata .section-header {
  display: none; /* Ingen collapse button */
}
```

Övriga sektioner: Collapsible med smart expansion

```
<Section
  id="lager-inkop"
  title="Lager & Inköp"
  isExpanded={expandedSections.includes('lager-inkop')}
  onToggle={() => toggleSection('lager-inkop')}
  fieldCount={{ filled: 3, total: 5 }}
  isPinned={pinnedSections.includes('lager-inkop')}
  onPin={() => togglePin('lager-inkop')}
>
  <FieldGrid>
    { /* Fields here */ }
  </FieldGrid>
</Section>
```

Section Header (collapsible):

```

<div className="section-header">
  <div className="section-header__left">
    {isPinned && <PinIcon />}
    <h2>{title}</h2>
    <Badge filled={3} total={5} />
  </div>
  <div className="section-header__right">
    <PinButton onClick={onPin} isPinned={isPinned} />
    <CollapseButton isExpanded={isExpanded} onClick={onToggle} />
  </div>
</div>

```

Styling:

```

.section-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 16px 24px;
  background: #FFFFFF;
  border: 1px solid #E0E0E0;
  border-radius: 8px;
  cursor: pointer;
  transition: all 0.2s ease;
}

.section-header:hover {
  background: #F5F5F5;
  border-color: #BDBDBD;
}

.section-header h2 {
  font-size: 18px;
  font-weight: 600;
  color: #212121;
  margin: 0;
}

/* Expanded state */
.section--expanded .section-header {
  border-bottom-left-radius: 0;
  border-bottom-right-radius: 0;
  border-bottom: none;
}

.section-content {
  max-height: 0;
  overflow: hidden;
  transition: max-height 0.3s ease;
}

.section--expanded .section-content {
  max-height: 5000px; /* Large enough value */
  border: 1px solid #E0E0E0;
  border-top: none;
  border-bottom-left-radius: 8px;
  border-bottom-right-radius: 8px;
  padding: 24px;
}

```


Badge-system för antal ifyllda fält

```
const Badge = ({ filled, total }) => {
  const percentage = (filled / total) * 100;
  const variant = filled === 0 ? 'empty' :
    filled === total ? 'complete' :
    'partial';

  return (
    <span className={`field-badge field-badge--${variant}`}>
      {filled}/{total}
    </span>
  );
};
```

Styling:

```
.field-badge {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  padding: 4px 10px;
  border-radius: 12px;
  font-size: 12px;
  font-weight: 600;
  margin-left: 12px;
}

.field-badge--empty {
  background: #F5F5F5;
  color: #757575;
}

.field-badge--partial {
  background: #FFE0B2;
  color: #E65100;
}

.field-badge--complete {
  background: #C8E6C9;
  color: #2E7D32;
}
```

Pin-funktionalitet för viktiga sektioner

Pin Button:

```
const PinButton = ({ isPinned, onClick }) => (
  <button
    className={`pin-button ${isPinned ? 'pin-button--active' : ''}`}
    onClick={(e) => {
      e.stopPropagation(); // Prevent section toggle
      onClick();
    }}
    title={isPinned ? 'Avpinna sektion' : 'Pinna sektion'}
  >
    {isPinned ? '📌' : '📍'}
  </button>
);
```

Styling:

```
.pin-button {
  background: transparent;
  border: 1px solid #E0E0E0;
  border-radius: 4px;
  padding: 6px 10px;
  cursor: pointer;
  font-size: 16px;
  transition: all 0.2s ease;
}

.pin-button:hover {
  background: #F5F5F5;
  border-color: #BDBDBD;
}

.pin-button--active {
  background: #FFF9C4;
  border-color: #FFE082;
}
```

Behavior:

- Pinnade sektioner expanderas automatiskt vid page load
 - Pinnade sektioner sorteras högst upp i listan (efter Grunddata)
 - Max 3 pinnade sektioner tillåtna (visa warning vid försök att pinna fler)
-

5. Interaktionsbeteenden

Klick på sidebar-länk → smooth scroll + auto-expand

```
const handleSectionClick = (sectionId) => {
  // 1. Expand section if collapsed
  if (!expandedSections.includes(sectionId)) {
    setExpandedSections([...expandedSections, sectionId]);
  }

  // 2. Wait for expansion animation (300ms)
  setTimeout(() => {
    // 3. Smooth scroll to section
    const element = document.getElementById(sectionId);
    if (element) {
      element.scrollIntoView({
        behavior: 'smooth',
        block: 'start',
        inline: 'nearest'
      });

      // 4. Offset for sticky header (if needed)
      const yOffset = -100; // 80px header + 20px padding
      const y = element.getBoundingClientRect().top + window.pageYOffset + yOffset;
      window.scrollTo({ top: y, behavior: 'smooth' });
    }
  }, 300);

  // 5. Update active section
  setActiveSection(sectionId);
};
```

State management för expanderade/collapsed sektioner

React State Structure:

```

const [sectionStates, setSectionStates] = useState({
  expandedSections: ['grunddata'], // Always include grunddata
  pinnedSections: [],
  activeSection: 'grunddata',
  fieldData: {} // Form data för varje field
});

// Toggle expansion
const toggleSection = (sectionId) => {
  setSectionStates(prev => ({
    ...prev,
    expandedSections: prev.expandedSections.includes(sectionId)
      ? prev.expandedSections.filter(id => id !== sectionId)
      : [...prev.expandedSections, sectionId]
  }));
};

// Toggle pin
const togglePin = (sectionId) => {
  setSectionStates(prev => {
    const isPinned = prev.pinnedSections.includes(sectionId);

    // Max 3 pinned sections
    if (!isPinned && prev.pinnedSections.length >= 3) {
      alert('Max 3 sektioner kan pinnas åt gången');
      return prev;
    }

    return {
      ...prev,
      pinnedSections: isPinned
        ? prev.pinnedSections.filter(id => id !== sectionId)
        : [...prev.pinnedSections, sectionId],
      // Auto-expand pinned sections
      expandedSections: isPinned
        ? prev.expandedSections
        : [...prev.expandedSections, sectionId]
    };
  });
};

```

LocalStorage för att komma ihåg användarens preferenser

```
// Storage key (unique per article)
const STORAGE_KEY = `traivo_article_${articleId}_layout_prefs`;

// Save preferences
useEffect(() => {
  const preferences = {
    expandedSections: sectionStates.expandedSections,
    pinnedSections: sectionStates.pinnedSections,
    timestamp: Date.now()
  };

  localStorage.setItem(STORAGE_KEY, JSON.stringify(preferences));
}, [sectionStates.expandedSections, sectionStates.pinnedSections]);

// Load preferences on mount
useEffect(() => {
  const stored = localStorage.getItem(STORAGE_KEY);

  if (stored) {
    try {
      const preferences = JSON.parse(stored);

      // Check if preferences are not too old (7 days)
      const isRecent = Date.now() - preferences.timestamp < 7 * 24 * 60 * 60 * 1000;

      if (isRecent) {
        setSectionStates(prev => ({
          ...prev,
          expandedSections: [...new Set([
            'grunddata', // Always include
            ...preferences.expandedSections,
            ...preferences.pinnedSections // Always expand pinned
          ])],
          pinnedSections: preferences.pinnedSections
        }));
      }
    } catch (error) {
      console.error('Failed to load layout preferences:', error);
    }
  }
}, [articleId]);
```

Keyboard shortcuts

```
useEffect(() => {
  const handleKeyPress = (e) => {
    // Ignore if user is typing in input field
    if (e.target.tagName === 'INPUT' || e.target.tagName === 'TEXTAREA') {
      return;
    }

    const key = e.key;
    const isCmdOrCtrl = e.metaKey || e.ctrlKey;

    // Cmd/Ctrl + 1-9: Jump to section
    if (isCmdOrCtrl && key >= '1' && key <= '9') {
      e.preventDefault();
      const index = parseInt(key) - 1;
      const sections = getAllSectionIds();
      if (sections[index]) {
        handleSectionClick(sections[index]);
      }
    }

    // Space: Toggle current section
    if (key === ' ' && activeSection) {
      e.preventDefault();
      toggleSection(activeSection);
    }

    // Arrow Up/Down: Navigate between sections
    if (key === 'ArrowDown' || key === 'ArrowUp') {
      e.preventDefault();
      const sections = getAllSectionIds();
      const currentIndex = sections.indexOf(activeSection);
      const nextIndex = key === 'ArrowDown'
        ? Math.min(currentIndex + 1, sections.length - 1)
        : Math.max(currentIndex - 1, 0);

      handleSectionClick(sections[nextIndex]);
    }

    // E: Expand all sections
    if (key === 'e' && !isCmdOrCtrl) {
      setSectionStates(prev => ({
        ...prev,
        expandedSections: getAllSectionIds()
      }));
    }

    // C: Collapse all (except grunddata and pinned)
    if (key === 'c' && !isCmdOrCtrl) {
      setSectionStates(prev => ({
        ...prev,
        expandedSections: ['grunddata', ...prev.pinnedSections]
      }));
    }
  };

  document.addEventListener('keydown', handleKeyPress);
  return () => document.removeEventListener('keydown', handleKeyPress);
}, [activeSection, sectionStates]);
```

Keyboard Shortcuts Summary:

- Cmd/Ctrl + 1-9 : Hoppa till sektion 1-9
- Space : Expandera/kollapsa aktiv sektion
- Arrow Up/Down : Navigera mellan sektioner
- E : Expandera alla sektioner
- C : Kollapsa alla (förutom grunddata och pinnade)

6. Tekniska komponenter

React komponentstruktur

```

src/
├── components/
│   ├── ArticleView/
│   │   ├── ArticleView.jsx           # Main container
│   │   ├── QuickNav/
│   │   │   ├── QuickNav.jsx         # Sidebar navigation
│   │   │   ├── SectionLink.jsx      # Individual nav link
│   │   │   └── quicknav.module.css
│   │   ├── MainContent/
│   │   │   ├── MainContent.jsx      # Content area container
│   │   │   ├── Section.jsx          # Collapsible section
│   │   │   ├── SectionHeader.jsx    # Header with badge & pin
│   │   │   ├── FieldGrid.jsx        # Grid layout for fields
│   │   │   └── maincontent.module.css
│   │   └── articleview.module.css
│   ├── shared/
│   │   ├── Badge.jsx               # Reusable badge component
│   │   ├── PinButton.jsx           # Pin/unpin button
│   │   └── CollapseButton.jsx       # Expand/collapse icon
│   └── ...
├── hooks/
│   ├── useSectionState.js          # Section expansion state
│   ├── useScrollSync.js            # Intersection Observer logic
│   ├── useKeyboardShortcuts.js     # Keyboard navigation
│   └── useLocalStoragePrefs.js     # Persistence logic
├── utils/
│   ├── sectionHelpers.js           # Field counting, validation
│   └── scrollHelpers.js             # Smooth scroll utilities
└── ...

```

Komponent-exempel: ArticleView.jsx


```

import React, { useState, useEffect } from 'react';
import QuickNav from '../QuickNav/QuickNav';
import MainContent from '../MainContent/MainContent';
import useSectionState from '../../hooks/useSectionState';
import useScrollSync from '../../hooks/useScrollSync';
import useKeyboardShortcuts from '../../hooks/useKeyboardShortcuts';
import styles from './articleview.module.css';

const ArticleView = ({ articleId, articleData }) => {
  const {
    expandedSections,
    pinnedSections,
    activeSection,
    toggleSection,
    togglePin,
    setActiveSection
  } = useSectionState(articleId);

  useScrollSync(setActiveSection);
  useKeyboardShortcuts({ activeSection, toggleSection });

  const sections = [
    { id: 'grunddata', title: 'Grunddata', alwaysExpanded: true },
    { id: 'lager-inkop', title: 'Lager & Inköp' },
    { id: 'ekonomi', title: 'Ekonomi' },
    { id: 'leverantor', title: 'Leverantör' },
    { id: 'tekniska-data', title: 'Tekniska data' },
    { id: 'ehandel', title: 'E-handel' },
    { id: 'bilder', title: 'Bilder & Media' },
    { id: 'dokument', title: 'Dokument' },
    { id: 'historik', title: 'Historik' }
  ];

  return (
    <div className={styles.articleView}>
      <QuickNav
        sections={sections}
        expandedSections={expandedSections}
        pinnedSections={pinnedSections}
        activeSection={activeSection}
        articleData={articleData}
        onSectionClick={(id) => {
          if (!expandedSections.includes(id)) {
            toggleSection(id);
          }
          // Scroll handled by SectionLink component
        }}
      />

      <MainContent
        sections={sections}
        expandedSections={expandedSections}
        pinnedSections={pinnedSections}
        articleData={articleData}
        onToggleSection={toggleSection}
        onTogglePin={togglePin}
      />
    </div>
  );
};

export default ArticleView;

```

State management approach

Custom Hook: `useSectionState.js`

```

import { useState, useEffect } from 'react';

const STORAGE_KEY_PREFIX = 'traivo_article_layout_';

export default function useSectionState(articleId) {
  const storageKey = `${STORAGE_KEY_PREFIX}${articleId}`;

  const [state, setState] = useState({
    expandedSections: ['grunddata'],
    pinnedSections: [],
    activeSection: 'grunddata'
  });

  // Load from localStorage on mount
  useEffect(() => {
    const stored = localStorage.getItem(storageKey);
    if (stored) {
      try {
        const parsed = JSON.parse(stored);
        const age = Date.now() - parsed.timestamp;

        // Only use if less than 7 days old
        if (age < 7 * 24 * 60 * 60 * 1000) {
          setState(prev => ({
            ...prev,
            expandedSections: [
              'grunddata',
              ...parsed.expandedSections,
              ...parsed.pinnedSections
            ].filter((v, i, a) => a.indexOf(v) === i), // unique
            pinnedSections: parsed.pinnedSections
          }));
        }
      } catch (e) {
        console.error('Failed to parse stored preferences', e);
      }
    }
  }, [articleId]);

  // Save to localStorage on change
  useEffect(() => {
    const toStore = {
      expandedSections: state.expandedSections.filter(id => id !== 'grunddata'),
      pinnedSections: state.pinnedSections,
      timestamp: Date.now()
    };
    localStorage.setItem(storageKey, JSON.stringify(toStore));
  }, [state.expandedSections, state.pinnedSections, storageKey]);

  const toggleSection = (sectionId) => {
    setState(prev => ({
      ...prev,
      expandedSections: prev.expandedSections.includes(sectionId)
        ? prev.expandedSections.filter(id => id !== sectionId && id !== 'grunddata')
        : [...prev.expandedSections, sectionId]
    }));
  };

  const togglePin = (sectionId) => {
    setState(prev => {
      const isPinned = prev.pinnedSections.includes(sectionId);

```

```

    if (!isPinned && prev.pinnedSections.length >= 3) {
      alert('Maximalt 3 sektioner kan pinnas');
      return prev;
    }

    const newPinned = isPinned
      ? prev.pinnedSections.filter(id => id !== sectionId)
      : [...prev.pinnedSections, sectionId];

    return {
      ...prev,
      pinnedSections: newPinned,
      expandedSections: isPinned
        ? prev.expandedSections
        : [...prev.expandedSections, sectionId]
    };
  });
};

const setActiveSection = (sectionId) => {
  setState(prev => ({ ...prev, activeSection: sectionId }));
};

return {
  expandedSections: state.expandedSections,
  pinnedSections: state.pinnedSections,
  activeSection: state.activeSection,
  toggleSection,
  togglePin,
  setActiveSection
};
}

```

CSS-implementation (Flexbox)

```

/* articleview.module.css */
.articleView {
  display: flex;
  flex-direction: row;
  width: 100%;
  min-height: calc(100vh - 80px); /* Subtract header height */
  background: #F8F9FA;
}

/* quicknav.module.css */
.quickNav {
  flex: 0 0 15%;
  min-width: 240px;
  max-width: 320px;
  position: sticky;
  top: 80px;
  height: calc(100vh - 80px);
  overflow-y: auto;
  background: #F8F9FA;
  border-right: 1px solid #E0E0E0;
  padding: 24px 16px;
  z-index: 100;
}

/* maincontent.module.css */
.mainContent {
  flex: 1;
  padding: 32px 48px;
  background: #FFFFFF;
  max-width: 1400px; /* Prevent too wide reading lines */
}

.section {
  margin-bottom: 24px;
}

.sectionHeader {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 16px 24px;
  background: #FFFFFF;
  border: 1px solid #E0E0E0;
  border-radius: 8px;
  cursor: pointer;
  transition: all 0.2s ease-in-out;
}

.sectionHeader:hover {
  background: #F5F5F5;
  border-color: #BDBDBD;
}

.sectionContent {
  max-height: 0;
  overflow: hidden;
  opacity: 0;
  transition: max-height 0.3s ease-in-out, opacity 0.3s ease-in-out;
}

.section.expanded .sectionContent {
  max-height: 5000px;
}

```

```

opacity: 1;
border: 1px solid #E0E0E0;
border-top: none;
border-bottom-left-radius: 8px;
border-bottom-right-radius: 8px;
padding: 24px;
}

.section.expanded .sectionHeader {
border-bottom-left-radius: 0;
border-bottom-right-radius: 0;
}

```

Intersection Observer för scroll-syncing

Custom Hook: useScrollSync.js

```

import { useEffect } from 'react';

export default function useScrollSync(setActiveSection) {
  useEffect(() => {
    // Find all section elements
    const sectionElements = document.querySelectorAll('[data-section-id]');

    if (sectionElements.length === 0) return;

    // Create intersection observer
    const observer = new IntersectionObserver(
      (entries) => {
        // Find the section with highest intersection ratio
        let maxRatio = 0;
        let activeId = null;

        entries.forEach(entry => {
          if (entry.intersectionRatio > maxRatio) {
            maxRatio = entry.intersectionRatio;
            activeId = entry.target.getAttribute('data-section-id');
          }
        });

        if (activeId && maxRatio > 0.3) {
          setActiveSection(activeId);
        }
      },
      {
        root: null,
        rootMargin: '-100px 0px -50% 0px', // Offset for header + trigger earlier
        threshold: [0, 0.1, 0.3, 0.5, 0.7, 1.0]
      }
    );

    // Observe all sections
    sectionElements.forEach(el => observer.observe(el));

    // Cleanup
    return () => {
      sectionElements.forEach(el => observer.unobserve(el));
    };
  }, [setActiveSection]);
}

```

Usage in Section component:

```
<div
  className={styles.section}
  data-section-id={id}
  id={id}
>
  {/* Section content */}
</div>
```

7. Responsivitet

Breakpoints för olika skärmstorlekar

```
/* Design Tokens */
:root {
  --breakpoint-mobile: 768px;
  --breakpoint-tablet: 1024px;
  --breakpoint-desktop: 1440px;
  --breakpoint-large-desktop: 1920px;
  --breakpoint-ultrawide: 2560px;
}
```

Layout per breakpoint

≥1920px (27"+ skärmar): Full Hybrid Layout

```
@media (min-width: 1920px) {
  .articleView {
    display: flex;
  }

  .quickNav {
    display: block;
    flex: 0 0 15%;
  }

  .mainContent {
    flex: 1;
    max-width: 1400px;
  }
}
```

1440px - 1919px (Laptop/Desktop): Hybrid med smalare sidebar


```

@media (min-width: 1440px) and (max-width: 1919px) {
  .quickNav {
    flex: 0 0 18%;
    min-width: 220px;
  }

  .mainContent {
    flex: 1;
    max-width: 1200px;
  }
}

```

1024px - 1439px (<27"): Tabs istället för sidebar

```

@media (min-width: 1024px) and (max-width: 1439px) {
  .articleView {
    display: block; /* Stack vertically */
  }

  .quickNav {
    position: static;
    height: auto;
    width: 100%;
    display: flex;
    flex-direction: row;
    overflow-x: auto;
    border-right: none;
    border-bottom: 2px solid #E0E0E0;
    padding: 12px 24px;
  }

  .sectionLink {
    flex: 0 0 auto;
    white-space: nowrap;
    margin-right: 8px;
  }

  .mainContent {
    width: 100%;
    max-width: 100%;
  }
}

```

≤1023px (Tablet): Fallback till klassisk accordion

```

@media (max-width: 1023px) {
  .quickNav {
    display: none; /* Hide sidebar completely */
  }

  .mainContent {
    width: 100%;
    padding: 16px;
  }

  /* All sections become accordion */
  .section {
    margin-bottom: 16px;
  }
}

```

≤767px (Mobile): Ej relevant för Traivo

```

@media (max-width: 767px) {
  /* Traivo är desktop-first, men lägg till basic mobile support */
  .mainContent {
    padding: 12px;
  }

  .sectionHeader {
    padding: 12px 16px;
  }

  .fieldBadge {
    font-size: 10px;
  }
}

```

Responsive behavior summary

Screen Size	Layout	Quick Nav	Sections	Notes
≥1920px	Hybrid	Sticky sidebar (15%)	Smart collapse	Primary target
1440-1919px	Hybrid	Sticky sidebar (18%)	Smart collapse	Laptop/small desktop
1024-1439px	Tabs	Horizontal tabs	Smart collapse	Fallback 1
≤1023px	Accordion	Hidden	All collapsible	Fallback 2
≤767px	Accordion	Hidden	All collapsible	Basic mobile (low priority)

8. Accessibility

ARIA-labels

```
// QuickNav
<nav
  className={styles.quickNav}
  aria-label="Artikel sektioner navigation"
  role="navigation"
>
  <ul role="list">
    <li>
      <a
        href={`#${section.id}`}
        role="link"
        aria-current={isActive ? 'true' : 'false'}
        aria-label={` ${section.title}, ${getStatusText(section.status)}, ${section.badge}`}
      >
        {/* Link content */}
      </a>
    </li>
  </ul>
</nav>

// Section
<section
  id={id}
  aria-labelledby={` ${id}-heading`}
  aria-expanded={isExpanded}
>
  <header
    id={` ${id}-heading`}
    role="button"
    aria-controls={` ${id}-content`}
    aria-expanded={isExpanded}
    tabIndex={0}
  >
    <h2>{title}</h2>
  </header>

  <div
    id={` ${id}-content`}
    role="region"
    aria-labelledby={` ${id}-heading`}
    hidden={!isExpanded}
  >
    {/* Content */}
  </div>
</section>

// Pin button
<button
  aria-label={isPinned ? 'Avpinna sektion' : 'Pinna sektion'}
  aria-pressed={isPinned}
>
  {isPinned ? '📌' : '📌'}
</button>
```

Keyboard navigation

Focus Management:

```
// Auto-focus first field when section expands
useEffect(() => {
  if (isExpanded && !wasExpanded) {
    const firstInput = sectionRef.current?.querySelector('input, textarea, select');
    if (firstInput && document.activeElement?.tagName !== 'INPUT') {
      // Small delay to ensure animation completes
      setTimeout(() => firstInput.focus(), 350);
    }
  }
}, [isExpanded]);

// Focus trap for modal-like interactions (if needed)
const trapFocus = (element) => {
  const focusableElements = element.querySelectorAll(
    'a[href], button, textarea, input, select, [tabindex]:not([tabindex="-1"])'
  );

  const firstElement = focusableElements[0];
  const lastElement = focusableElements[focusableElements.length - 1];

  element.addEventListener('keydown', (e) => {
    if (e.key === 'Tab') {
      if (e.shiftKey && document.activeElement === firstElement) {
        e.preventDefault();
        lastElement.focus();
      } else if (!e.shiftKey && document.activeElement === lastElement) {
        e.preventDefault();
        firstElement.focus();
      }
    }
  });
};
```

Keyboard Shortcuts Legend:

```
// Add help modal/tooltip
const KeyboardShortcutsHelp = () => (
  <div className={styles.shortcutsHelp} role="dialog" aria-label="Tangentbordsgen-
vägar">
    <h3>Tangentbordsgenvägar</h3>
    <dl>
      <dt><kbd>Cmd/Ctrl</kbd> + <kbd>1-9</kbd></dt>
      <dd>Hoppa till sektion 1-9</dd>

      <dt><kbd>↑</kbd> / <kbd>↓</kbd></dt>
      <dd>Navigera mellan sektioner</dd>

      <dt><kbd>Space</kbd></dt>
      <dd>Expandera/kollapsa aktiv sektion</dd>

      <dt><kbd>E</kbd></dt>
      <dd>Expandera alla sektioner</dd>

      <dt><kbd>C</kbd></dt>
      <dd>Kollapsa alla sektioner</dd>

      <dt><kbd>?</kbd></dt>
      <dd>Visa denna hjälp</dd>
    </dl>
  </div>
);
```

Screen reader considerations

Live Regions for status updates:

```
const [announcement, setAnnouncement] = useState('');

// Announce when section expands/collapses
const handleToggle = (sectionId) => {
  toggleSection(sectionId);
  const section = sections.find(s => s.id === sectionId);
  const isExpanding = !expandedSections.includes(sectionId);

  setAnnouncement(
    isExpanding
      ? `${section.title} expanderad`
      : `${section.title} kollapsad`
  );
};

// Live region
<div
  role="status"
  aria-live="polite"
  aria-atomic="true"
  className="sr-only"
>
  {announcement}
</div>
```

Screen-reader only text:

```
.sr-only {
  position: absolute;
  width: 1px;
  height: 1px;
  padding: 0;
  margin: -1px;
  overflow: hidden;
  clip: rect(0, 0, 0, 0);
  white-space: nowrap;
  border-width: 0;
}
```

Semantic HTML:

- Använd `<nav>` för Quick Nav
- Använd `<section>` för artikelsektioner
- Använd `<header>` för section headers
- Använd `<button>` för interaktiva element (inte `<div>` med `onClick`)
- Använd `<a>` för navigation links

9. Implementation Priority

Phase 1: Basic Sidebar + Scroll Navigation (Week 1)

Mål: Få grundläggande hybrid layout att fungera

- [] Skapa flexbox layout med 15/85 split
- [] Implementera sticky sidebar
- [] Skapa SectionLink komponenter med basic styling
- [] Implementera scroll-to-section funktionalitet
- [] Implementera Intersection Observer för active state syncing
- [] Basic responsivitet (hide sidebar på <1440px)

Deliverables:

- Användare kan klicka på sidebar link och scrolla till sektion
- Sidebar highlightar aktiv sektion vid scrolling
- Layout fungerar på desktop ($\geq 1440\text{px}$)

Acceptance Criteria:

- ✓ Sidebar är sticky och följer med vid scroll
- ✓ Klick på sidebar link scollar smooth till rätt sektion
- ✓ Active state uppdateras korrekt vid scroll
- ✓ Ingen horizontal scroll
- ✓ Layout är responsiv för desktop

Phase 2: Smart Expansion + Badges (Week 2)

Mål: Lägg till collapse/expand funktionalitet och status badges

- [] Implementera Section komponenter med collapse/expand
- [] Skapa useSectionState hook för state management

- [] Implementera badge system (filled/total field counter)
- [] Lägg till status icons (✓, △, ○)
- [] Implementera auto-expand när sidebar link klickas
- [] Lägg till smooth transitions för expand/collapse

Deliverables:

- Sektioner kan expanderas/kollapsas
- Badges visar antal ifyllda fält
- Status icons visar section completeness
- Sidebar click expanderar automatiskt sektion

Acceptance Criteria:

- ✓ Alla sektioner (förutom Grunddata) är collapsible
 - ✓ Grunddata är alltid expanderad
 - ✓ Badges visar korrekt antal (filled/total)
 - ✓ Status icons matchar section state
 - ✓ Expand animation är smooth (300ms)
 - ✓ Klick på sidebar både scollar och expanderar
-

Phase 3: Pin Functionality + Keyboard Shortcuts (Week 3)

Mål: Lägg till pin-funktionalitet och keyboard navigation

- [] Implementera pin/unpin button i section headers
- [] Lägg till pin indicator i sidebar
- [] Implementera max 3 pinned sections logic
- [] Auto-expand pinned sections on load
- [] Implementera keyboard shortcuts (Cmd+1-9, Space, E, C, arrows)
- [] Skapa useKeyboardShortcuts hook

Deliverables:

- Användare kan pinna upp till 3 sektioner
- Pinnade sektioner expanderas automatiskt
- Keyboard shortcuts fungerar

Acceptance Criteria:

- ✓ Pin button finns i alla section headers
 - ✓ Max 3 sektioner kan pinnas (visa varning vid överskridning)
 - ✓ Pinnade sektioner expanderas vid page load
 - ✓ Keyboard shortcuts fungerar korrekt
 - ✓ Pin indicator visas i sidebar
-

Phase 4: State Persistence + Polish (Week 4)

Mål: Spara user preferences och finputs UX

- [] Implementera localStorage persistence för expanded/pinned state
- [] Lägg till 7-dagars expiration för stored preferences
- [] Implementera accessibility features (ARIA labels, keyboard nav)

- [] Lägg till keyboard shortcuts help modal (?)
- [] Polish transitions och animations
- [] Implementera live regions för screen readers
- [] Performance optimization (lazy rendering, virtualization if needed)

Deliverables:

- User preferences sparas mellan sessioner
- Full accessibility support
- Polished UX med smooth animations

Acceptance Criteria:

- ✓ Expanded/pinned state sparas i localStorage
- ✓ State återställs korrekt vid page reload
- ✓ Gamla preferences (>7 dagar) rensas
- ✓ ARIA labels på alla interaktiva element
- ✓ Screen reader support fungerar korrekt
- ✓ Inga accessibility errors i Axe/Lighthouse

10. Acceptanskriterier

Funktionella krav

Navigation:

- [] Sidebar visar alla sektioner i artikelvyn
- [] Klick på sidebar link scollar till och expanderar sektion
- [] Smooth scroll med korrekt offset för header
- [] Active section highlightas i sidebar vid scroll
- [] Sidebar är sticky och följer med vid vertical scroll

Section Management:

- [] Grunddata är alltid expanderad och kan ej kollapsas
- [] Övriga sektioner kan expanderas/kollapsas
- [] Expand/collapse animation är smooth (300ms)
- [] Endast en animation åt gången (ingen queue)

Status & Badges:

- [] Status icons (✓/△/○) reflekterar section completeness
- [] Badges visar korrekt antal ifyllda/totala fält
- [] Badge färger matchar status (grön/orange/grå)

Pin Functionality:

- [] Användare kan pinna/avpinna sektioner
- [] Max 3 sektioner kan vara pinnade samtidigt
- [] Pinnade sektioner expanderas automatiskt vid load
- [] Pin indicator visas i både sidebar och section header

State Persistence:

- [] Expanded/collapsed state sparas i localStorage
- [] Pinned sections sparas i localStorage
- [] State återställs korrekt vid page reload

- [] State är artikel-specifikt (per articleId)
- [] Gamla preferences (>7 dagar) ignoreras

Keyboard Navigation:

- [] Cmd/Ctrl + 1-9 hoppar till section
- [] Arrow up/down navigerar mellan sections
- [] Space expanderar/kollapsar aktiv section
- [] E expanderar alla sections
- [] C kollapsar alla (förutom grunddata & pinned)

Tekniska krav**Performance:**

- [] Initial render < 1s
- [] Scroll performance 60fps
- [] Expand/collapse animation 60fps
- [] Inga memory leaks (verify med DevTools)

Responsivitet:

- [] Layout fungerar på ≥ 1920 px (primary target)
- [] Layout fungerar på 1440-1919px
- [] Fallback till tabs/accordion på <1440px
- [] Inga horizontal scrollbars

Accessibility:

- [] ARIA labels på alla interaktiva element
- [] Keyboard navigation fungerar fullständigt
- [] Focus states tydligt synliga
- [] Screen reader announcements vid state changes
- [] Axe/Lighthouse accessibility score >90

Browser Support:

- [] Chrome/Edge (latest)
- [] Firefox (latest)
- [] Safari (latest)

Design krav**Visual Polish:**

- [] Spacing följer design tokens
- [] Färger matchar Traivo design system
- [] Hover states på alla interaktiva element
- [] Active states tydligt synliga
- [] Transitions smooth och consistent

Typography:

- [] Font sizes följer design system
 - [] Line heights optimerade för läsbarhet
 - [] Font weights korrekt använt
-

11. Design Tokens

Färgpalett

```
:root {  
  /* Primary Colors */  
  --color-primary: #2196F3;          /* Traivo Blue */  
  --color-primary-light: #64B5F6;  
  --color-primary-dark: #1976D2;  
  
  /* Status Colors */  
  --color-success: #4CAF50;          /* Complete */  
  --color-warning: #FF9800;         /* Partial */  
  --color-error: #F44336;  
  --color-info: #2196F3;  
  
  /* Neutral Colors */  
  --color-gray-50: #FAFAFA;  
  --color-gray-100: #F5F5F5;  
  --color-gray-200: #E0E0E0;  
  --color-gray-300: #BDBDBD;  
  --color-gray-400: #9E9E9E;  
  --color-gray-500: #757575;  
  --color-gray-600: #616161;  
  --color-gray-700: #424242;  
  --color-gray-800: #212121;  
  
  /* Background Colors */  
  --bg-primary: #FFFFFF;  
  --bg-secondary: #F8F9FA;  
  --bg-tertiary: #F5F5F5;  
  
  /* Border Colors */  
  --border-light: #E0E0E0;  
  --border-medium: #BDBDBD;  
  --border-dark: #9E9E9E;  
  
  /* State-specific Backgrounds */  
  --bg-hover: #F5F5F5;  
  --bg-active: #E3F2FD;  
  --bg-pinned: #FFF9C4;  
  
  /* Badge Colors */  
  --badge-empty-bg: #F5F5F5;  
  --badge-empty-text: #757575;  
  --badge-partial-bg: #FFE0B2;  
  --badge-partial-text: #E65100;  
  --badge-complete-bg: #C8E6C9;  
  --badge-complete-text: #2E7D32;  
}
```

Typography

```
:root {
  /* Font Families */
  --font-primary: 'Inter', -apple-system, BlinkMacSystemFont, 'Segoe UI', sans-serif;
  --font-mono: 'Fira Code', 'Courier New', monospace;

  /* Font Sizes */
  --font-size-xs: 10px;
  --font-size-sm: 12px;
  --font-size-base: 14px;
  --font-size-md: 16px;
  --font-size-lg: 18px;
  --font-size-xl: 20px;
  --font-size-2xl: 24px;
  --font-size-3xl: 30px;

  /* Font Weights */
  --font-weight-normal: 400;
  --font-weight-medium: 500;
  --font-weight-semibold: 600;
  --font-weight-bold: 700;

  /* Line Heights */
  --line-height-tight: 1.2;
  --line-height-normal: 1.5;
  --line-height-relaxed: 1.75;
}
```

Spacing System

```
:root {
  /* Spacing Scale (4px base) */
  --spacing-0: 0;
  --spacing-1: 4px;
  --spacing-2: 8px;
  --spacing-3: 12px;
  --spacing-4: 16px;
  --spacing-5: 20px;
  --spacing-6: 24px;
  --spacing-7: 28px;
  --spacing-8: 32px;
  --spacing-10: 40px;
  --spacing-12: 48px;
  --spacing-16: 64px;

  /* Semantic Spacing */
  --spacing-xs: var(--spacing-1); /* 4px */
  --spacing-sm: var(--spacing-2); /* 8px */
  --spacing-md: var(--spacing-4); /* 16px */
  --spacing-lg: var(--spacing-6); /* 24px */
  --spacing-xl: var(--spacing-8); /* 32px */
  --spacing-2xl: var(--spacing-12); /* 48px */
}
```

Border Radius

```
:root {
  --radius-sm: 4px;
  --radius-md: 6px;
  --radius-lg: 8px;
  --radius-xl: 12px;
  --radius-full: 9999px; /* Pills/badges */
}
```

Shadows

```
:root {
  --shadow-sm: 0 1px 2px 0 rgba(0, 0, 0, 0.05);
  --shadow-md: 0 4px 6px -1px rgba(0, 0, 0, 0.1);
  --shadow-lg: 0 10px 15px -3px rgba(0, 0, 0, 0.1);
  --shadow-xl: 0 20px 25px -5px rgba(0, 0, 0, 0.1);
}
```

Transitions

```
:root {
  /* Durations */
  --duration-fast: 150ms;
  --duration-normal: 200ms;
  --duration-slow: 300ms;

  /* Easing Functions */
  --ease-in: cubic-bezier(0.4, 0, 1, 1);
  --ease-out: cubic-bezier(0, 0, 0.2, 1);
  --ease-in-out: cubic-bezier(0.4, 0, 0.2, 1);

  /* Common Transitions */
  --transition-fast: all var(--duration-fast) var(--ease-out);
  --transition-normal: all var(--duration-normal) var(--ease-out);
  --transition-slow: all var(--duration-slow) var(--ease-in-out);
}
```

Z-index Scale

```
:root {
  --z-base: 1;
  --z-dropdown: 100;
  --z-sticky: 100; /* Sidebar */
  --z-header: 200;
  --z-overlay: 300;
  --z-modal: 400;
  --z-popover: 500;
  --z-tooltip: 600;
}
```

Usage Examples

Section Header:

```
.sectionHeader {
  padding: var(--spacing-md) var(--spacing-lg);
  background: var(--bg-primary);
  border: 1px solid var(--border-light);
  border-radius: var(--radius-lg);
  font-size: var(--font-size-lg);
  font-weight: var(--font-weight-semibold);
  transition: var(--transition-normal);
}

.sectionHeader:hover {
  background: var(--bg-hover);
  border-color: var(--border-medium);
}
```

Badge:

```
.badge {
  padding: var(--spacing-1) var(--spacing-2);
  border-radius: var(--radius-full);
  font-size: var(--font-size-sm);
  font-weight: var(--font-weight-semibold);
}

.badge--complete {
  background: var(--badge-complete-bg);
  color: var(--badge-complete-text);
}
```

Slutord

Detta dokument ger en komplett specifikation för att implementera Hybrid Layout i Traivos artikelvy. All information som behövs för att börja utveckla finns här:

- **Visuell design** med exakta mått
- **Komponentstruktur** med kodexempel
- **State management** approach
- **Interaktionsbeteenden** i detalj
- **Accessibility** requirements
- **Implementation roadmap** i 4 faser

Nästa steg:

1. Granska denna spec med teamet
2. Sätt up development environment
3. Starta med Phase 1 (Basic Sidebar + Scroll)
4. Iterera genom faserna

Frågor/Feedback:

Kontakta produktägare eller UX-team om något är oklart.

Version: 1.0

Senast uppdaterad: 2026-06-15

Författare: Traivo Product Team